INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon Web Services, VA -- station KHEF, America/New_York
Committed pair	OSM 1510517638 -> 1510517639 Amazon Web Services / Amazon Web Services
Facade gap	63.4 m between the two halls
Weather record	42,554 real hourly records from KHEF
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4054 C at 305 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-04-11 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.080	18.0	5.000	0.949
01	mechanical	yes	switch budget	4.996	18.0	3.893	1.277
02	mechanical	yes	switch budget	4.166	18.0	2.223	1.174
03	mechanical	yes	switch budget	4.446	18.0	2.783	1.132
04	mechanical	yes	switch budget	3.056	18.0	1.113	1.078

05	mechanical	yes	switch budget	1.691	18.0	0.003	1.044
06	mechanical	yes	switch budget	2.191	18.0	-0.607	1.349
07	mechanical	yes	switch budget	4.721	18.0	3.893	1.656
08	mechanical	yes	switch budget	8.611	18.0	8.893	2.385
09	mechanical	yes	switch budget	11.458	18.0	12.780	2.650
10	mechanical	yes	switch budget	16.111	18.0	16.673	2.941
11	mechanical	no	dry-bulb	18.960	18.0	20.000	2.118
12	mechanical	no	dry-bulb	21.391	18.0	22.783	1.771
13	mechanical	no	dry-bulb	23.054	18.0	23.890	1.437
14	mechanical	no	dry-bulb	24.439	18.0	24.440	1.323
15	mechanical	no	dry-bulb	25.705	18.0	25.561	1.105
16	mechanical	no	refusal	25.519	18.0	25.000	1.263
17	mechanical	no	dry-bulb	24.894	18.0	24.440	1.345
18	mechanical	no	refusal	24.377	18.0	22.780	1.675
19	mechanical	no	dry-bulb	21.661	18.0	18.333	1.887
20	mechanical	no	dry-bulb	20.001	18.0	15.563	2.000
21	FREE	yes	-	17.501	18.0	12.223	1.849
22	FREE	yes	-	14.996	18.0	11.113	1.730
23	FREE	yes	-	12.221	18.0	8.333	1.383

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.080 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.996 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.166 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.446 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.056 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.691 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.191 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.721 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.611 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.458 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.111 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.960 C against a 18.0 C limit, so it fails by 0.960 C. It would take a limit 0.960 C higher, or a bound 0.960 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.391 C against a 18.0 C limit, so it fails by 3.391 C. It would take a limit 3.391 C higher, or a bound 3.391 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.054 C against a 18.0 C limit, so it fails by 5.054 C. It would take a limit 5.054 C higher, or a bound 5.054 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.439 C against a 18.0 C limit, so it fails by 6.439 C. It would take a limit 6.439 C higher, or a bound 6.439 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.705 C against a 18.0 C limit, so it fails by 7.705 C. It would take a limit 7.705 C higher, or a bound 7.705 C tighter, to change this hour.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.894 C against a 18.0 C limit, so it fails by 6.894 C. It would take a limit 6.894 C higher, or a bound 6.894 C tighter, to change this hour.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.661 C against a 18.0 C limit, so it fails by 3.661 C. It would take a limit 3.661 C higher, or a bound 3.661 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.001 C against a 18.0 C limit, so it fails by 2.001 C. It would take a limit 2.001 C higher, or a bound 2.001 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.501 C, which is 0.499 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.849 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.2738 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.996 C, which is 3.004 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.730 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.2738 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.221 C, which is 5.779 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.383 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.2738 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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